MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions — MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The value of $\dfrac{\sin 30^\circ + \cos 60^\circ}{\tan 45^\circ}$ is: [1]

(A) 0 (B) 1 (C) $\frac{1}{2}$ (D) 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 1

Justification: $\dfrac{1/2 + 1/2}{1} = \dfrac{1}{1} = 1.$

Q2. If $\csc\theta = \frac{5}{3}$ and $\cot\phi = \frac{12}{5}$, where θ, ϕ are acute, then $\cos\theta \sin\phi$ is: [1]

(A) $\frac{4}{13}$ (B) $\frac{3}{13}$
(C) $\frac{4}{5}$ (D) $\frac{5}{13}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{4}{13}$

Justification: From $\csc\theta = \frac{5}{3} \implies \sin\theta = \frac{3}{5} \implies \cos\theta = \frac{4}{5}$. From $\cot\phi = \frac{12}{5}$ (5–12–13 right triangle), $\sin\phi = \frac{5}{13}$. Thus $\cos\theta \sin\phi = \frac{4}{5} \times \frac{5}{13} = \frac{4}{13}$.

Q3. If $2\sin^2\theta - \cos^2\theta = \frac{1}{2}$, then the value of acute angle θ is: [1]

(A) $30°$ (B) $60°$ (C) $45°$ (D) $90°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C)** $45°$

Justification: $2\sin^2\theta - (1 - \sin^2\theta) = \frac{1}{2} \implies 3\sin^2\theta = \frac{3}{2} \implies \sin^2\theta = \frac{1}{2} \implies \sin\theta = \frac{1}{\sqrt{2}} \implies \theta = 45°$.

Q4. If $\sec\theta + \tan\theta = p$, then the value of $\sin\theta$ in terms of p is: [1]

(A) $\frac{p^2-1}{p^2+1}$ (B) $\frac{p^2+1}{p^2-1}$

(C) $\frac{2p}{p^2+1}$ (D) $\frac{2p}{p^2-1}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $\frac{p^2-1}{p^2+1}$

Justification: Since $\sec\theta - \tan\theta = \frac{1}{p}$, subtracting and adding give $2\tan\theta = \frac{p^2-1}{p}$ and $2\sec\theta = \frac{p^2+1}{p}$. Dividing yields $\sin\theta = \frac{2\tan\theta}{2\sec\theta} = \frac{p^2-1}{p^2+1}$.

Q5. If $\sin\theta - \cos\theta = \frac{1}{2}$, then the value of $\sin\theta\cos\theta$ is: [1]

(A) $\frac{3}{8}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $\frac{3}{8}$

Justification: Squaring both sides: $\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = \frac{1}{4} \implies 1 - 2\sin\theta\cos\theta = \frac{1}{4} \implies 2\sin\theta\cos\theta = \frac{3}{4} \implies \sin\theta\cos\theta = \frac{3}{8}$.

Q6. Assertion (A): $\sec^2\theta + \csc^2\theta = \sec^2\theta\csc^2\theta$ for any acute angle θ.

Reason (R): $\dfrac{1}{\cos^2\theta} + \dfrac{1}{\sin^2\theta} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin^2\theta\cos^2\theta} = \dfrac{1}{\sin^2\theta\cos^2\theta} = \sec^2\theta\csc^2\theta.$ [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Justification: Both (A) and (R) are true, and (R) gives the exact step-by-step derivation of (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. If $\sqrt{3}\sin\theta - \cos\theta = 0$ and $0° < \theta < 90°$, find the value of θ. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]
Step 1: Form Equation in $\tan\theta$ [1 Mark]

$$\sqrt{3}\sin\theta = \cos\theta \implies \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{3}} \implies \tan\theta = \frac{1}{\sqrt{3}}$$

Step 2: Determine Acute Angle θ [1 Mark]

$$\tan\theta = \tan 30° \implies \boxed{\theta = \mathbf{30°}}$$

Q8. Prove that: $\dfrac{\cos A}{1-\tan A} + \dfrac{\sin A}{1-\cot A} = \sin A + \cos A.$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]
Step 1: Convert into $\sin A, \cos A$ & Common Denominator [1 Mark]

$$\text{LHS} = \frac{\cos^2 A}{\cos A - \sin A} + \frac{\sin^2 A}{\sin A - \cos A} = \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A}$$

Step 2: Factorize Numerator [1 Mark]

$$= \frac{(\cos A - \sin A)(\cos A + \sin A)}{\cos A - \sin A} = \boxed{\sin \mathbf{A} + \cos \mathbf{A} = \text{RHS} \quad [\text{Hence Proved}]}$$

Q9. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, prove that $m^2 - n^2 = 4\sqrt{mn}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Evaluate LHS $= m^2 - n^2$ [1.5 Marks]

$$m^2 - n^2 = (\tan\theta + \sin\theta)^2 - (\tan\theta - \sin\theta)^2 = 4\tan\theta\sin\theta$$

Step 2: Evaluate RHS $= 4\sqrt{mn}$ **(for acute** θ**,** $\tan\theta\sin\theta > 0$**)** [1.5 Marks]

$$mn = \tan^2\theta - \sin^2\theta = \sin^2\theta\left(\frac{1}{\cos^2\theta} - 1\right) = \sin^2\theta\left(\frac{1-\cos^2\theta}{\cos^2\theta}\right) = \tan^2\theta\sin^2\theta$$

$$4\sqrt{mn} = 4\sqrt{\tan^2\theta\sin^2\theta} = 4\tan\theta\sin\theta \implies \boxed{\mathbf{m^2 - n^2 = 4\sqrt{mn}} \quad \text{[Hence Proved]}}$$

Q10. Prove that: $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1 + \sin A}{\cos A}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Replace 1 **in Numerator Using** $\sec^2 A - \tan^2 A = 1$ [1.5 Marks]

$$\text{LHS} = \frac{(\tan A + \sec A) - (\sec^2 A - \tan^2 A)}{\tan A - \sec A + 1} = \frac{(\tan A + \sec A)[1 - (\sec A - \tan A)]}{\tan A - \sec A + 1}$$

Step 2: Cancel Common Factor & Convert to $\sin A, \cos A$ [1.5 Marks]

$$\text{LHS} = \frac{(\tan A + \sec A)(\tan A - \sec A + 1)}{\tan A - \sec A + 1} = \tan A + \sec A$$

$$= \frac{\sin A}{\cos A} + \frac{1}{\cos A} = \boxed{\frac{\mathbf{1 + \sin A}}{\mathbf{\cos A}} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q11. Prove that: $\dfrac{1}{\sec x - \tan x} - \dfrac{1}{\cos x} = \dfrac{1}{\cos x} - \dfrac{1}{\sec x + \tan x}$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Simplify LHS Using $\sec^2 x - \tan^2 x = 1$ [2 Marks]

$$\text{LHS} = \frac{\sec^2 x - \tan^2 x}{\sec x - \tan x} - \sec x = (\sec x + \tan x) - \sec x = \tan x$$

Step 2: Simplify RHS Analogously [2 Marks]

$$\text{RHS} = \sec x - \frac{\sec^2 x - \tan^2 x}{\sec x + \tan x} = \sec x - (\sec x - \tan x) = \tan x$$

Step 3: Conclusion [1 Mark]

$$\boxed{\therefore \text{LHS} = \text{RHS} = \tan\mathbf{x} \quad \text{[Hence Proved]}}$$

Q12. Prove that: $\dfrac{\sin A+\cos A}{\sin A-\cos A}+\dfrac{\sin A-\cos A}{\sin A+\cos A}=\dfrac{2}{\sin^2 A-\cos^2 A}=\dfrac{2}{2\sin^2 A-1}$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Combine Fractions over Common Denominator [2 Marks]

$$\text{LHS}=\frac{(\sin A+\cos A)^2+(\sin A-\cos A)^2}{(\sin A-\cos A)(\sin A+\cos A)}=\frac{2(\sin^2 A+\cos^2 A)}{\sin^2 A-\cos^2 A}$$

Step 2: Apply $\sin^2 A+\cos^2 A=1$ [1.5 Marks]

$$=\frac{2(1)}{\sin^2 A-\cos^2 A}=\frac{2}{\sin^2 A-\cos^2 A}$$

Step 3: Substitute $\cos^2 A=1-\sin^2 A$ [1.5 Marks]

$$=\frac{2}{\sin^2 A-(1-\sin^2 A)}=\boxed{\frac{\mathbf{2}}{\mathbf{2\sin^2 A-1}}\quad\text{[Hence Proved]}}$$

SECTION E: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. High-Mast Stadium Floodlight Luminaire Projection

A sports stadium installs high-mast lighting towers to illuminate the soccer field. A mast AB of vertical height 24 m supports a bank of floodlights at top point A. The central axis of the floodlight beam hits the ground at point C, making an angle of elevation θ from the ground such that $\sin\theta=\dfrac{3}{5}$. **Based on the above geometric model, answer the following questions:**

(i) Find the value of $\tan\theta$ and $\cos\theta$ for this beam angle. [1]

(ii) Calculate the ground distance BC from the base of the mast to the beam target point. [1]

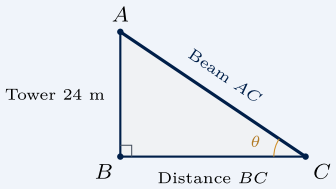
(iii) Find the direct beam throw distance AC from the luminaires to the ground target, and verify the identity $1+\cot^2\theta=\csc^2\theta$. [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): Values of $\cos\theta$ **and** $\tan\theta$ [1 Mark]

For acute θ with $\sin\theta=\frac{3}{5}$ (3–4–5 right triangle), $\cos\theta=\sqrt{1-\frac{9}{25}}=\frac{4}{5}$:

$$\boxed{\tan\theta=\frac{\mathbf{3}}{\mathbf{4}},\qquad\cos\theta=\frac{\mathbf{4}}{\mathbf{5}}}$$



Sub-part (ii): Ground Distance BC [1 Mark]

$$\tan\theta=\frac{AB}{BC}\implies\frac{3}{4}=\frac{24}{BC}\implies\boxed{BC=\mathbf{32}\text{ m}}$$

Sub-part (iii): Beam Throw AC **& Identity Verification** [2 Marks]

$$\sin\theta=\frac{AB}{AC}\implies\frac{3}{5}=\frac{24}{AC}\implies\boxed{AC=\mathbf{40}\text{ m}}$$

$$1+\cot^2\theta=1+\left(\frac{4}{3}\right)^2=1+\frac{16}{9}=\frac{25}{9}=\left(\frac{5}{3}\right)^2=\csc^2\theta\quad\text{(Verified)}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★